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Section A

"Software Engineering"

Assignment

Hospital Management System

Purpose:

The primary purpose of Hospital Management System is to digitalize, centralize and streamline hospital operations to improve patient care, reduce human error, and optimize clinical and administrative workflows.

Users:

doctors and Specialists: Review histories log diagnosis and write prescriptions.

Nurses and Ward staff: Monitor vitals administer medication, and update room statuses.

Receptionists and admins: Schedule appointments register patients, and manage staff shifts.

Patients: view medical history, book appointments.

Stakeholders:

Hospital boards and Executives are interested in operational efficiency, ROI and hospital reputation. Government and Health authorities

require strict regulatory compliance.

Functional Requirements:

Patient Registration: Create, search and update EHR.

Appointment Scheduling: Book, reschedule, and cancel appointments on a doctor's calendar.

e-prescriptions: Generate electronic prescriptions and sync them directly to the pharmacy.

Billing and insurance: Calculate patient costs and electronically process insurance claims.

Non-functional Requirements:

Security: Encrypt all patient data to meet healthcare laws.

Availability: Maintain 99.99% system uptime for constant emergency access.

Speed: Load critical patient profiles in under 2 seconds.

Risks:

Cyberattacks: Ransomware can lock medical files, delaying life-saving treatments.

System Crashes: Server downtime blocks access to critical allergies and histories.

Quality Attributes:

Reliability: Executes clinical orders perfectly without dropping or corrupting data.

Usability: Simple, fast screens that staff can use easily under stress.

AI use:

X-Ray/MRI Diagnosis: AI models ^{can scan} X-rays and MRIs to quickly spot tumors, fractures or anomalies.

Smart Triage: AI can analyze vitals to predict which emergency patients need urgent care first.

What could go wrong if AI is used incorrectly?

Flaw AI data can miss critical conditions like tumors, leading to fatal treatment delays.